

INGB417 FACILITIES DESIGN alignment		© Willemien Kruger Feb 2025	
ENGINEERING DESIGN PROCESS STEPS	FACILITIES DESIGN STEPS aligned (with Deliverables)	REPORT aligned (see 'template')	Evaluation criteria fit
		Cover page	
		1. Executive summary (write last)	
		2. Background to the project	
STEP 1 - DEFINE THE PROBLEM/OPPORTUNITY	Define the PROBLEM/OPPORTUNITY (for redesign) or NEED (for new design) and also Define OBJECTIVES (facility need to be able to support this)	3. Design Problem and Objectives	
	Define REQUIREMENTS STATEMENTS (facility need to do this&that, to measure against at end)	4. Design Requirements	
	Decide on Design ASSUMPTIONS made and what CONSTRAINTS are there before detail specifications	5. Design Assumptions and Constraints	
STEP 2 - ANALYZE THE PROBLEM &	Do a DETAILED ANALYSIS of SPECIFICATIONS (as part of Productio PROCESS DESIGN)	6. Detail Design Specifications [this is the biggest part of the report]	
STEP 3 - DETERMINE DESIGN REQUIREMENTS / SPECIFICATIONS	Get lots of DATA by asking the right questions [see Getting Data guidance document]	Sub paragraphs under chapter 6. or put it in Appendices	
	Apply MATHS to <i>do capacity and other Calculations</i> to get to the specific requirements in next steps as well as TOOLS and TECHNIQUES to get this data [see Getting Data guidance document]	Structure sub paragraphs as you want it and as it makes sense, under chapter 6.as it logically relates to data analysis or in Appendices	
	Determine overall SPACE REQUIREMENTS and for different 'departments'/functions/areas/etc [see Getting Data guidance document		
STEP 4 - DEVELOP ALTERNATIVE DESIGNS	Develop alternative BLOCK LAYOUTS to meet the requirements	7. Design Options/Alternatives	
	Can apply TOOLS at this point also for new facility [see Getting Data guidance document]	Sub headings under chapter 7 arranged as you want it to make sense.	
STEP 5 - EVALUATE ALTERNATIVES	Do EVALUATION by applying tools eg. Weighted criteria, Pros/Cons analysis	8. Evaluating Alternatives	
	Measure against requirements and/or specifications		
STEP 6 - SELECT A DESIGN and justify why	Draw the MASTER FACILITY PLAN even if its combination of alternatives		
STEP 7 - MFP and PRESENTATION	Draw the MASTER FACILITY PLAN and Develop a BUSINESS CASE [cost, risks, schedule estimates] to include in Report	9. Final Facility Design (MFP) and 10. Business Case	

clear interpretation of problem and formulation of design requirements

appropriate selection and effective use of analytical tools and techniques

practical insight shown in the design process and design solution

sufficient design detail provided to meet the design requirements

effective use of a structured and rigorous design process

effective communication of the design process and solution